

German University in Cairo (GUC)
Media Engineering and Technology (MET)

Automated Hallucination Detection Report

Neuro-Symbolic Verification via Knowledge Graphs

Bachelor Thesis Project Output

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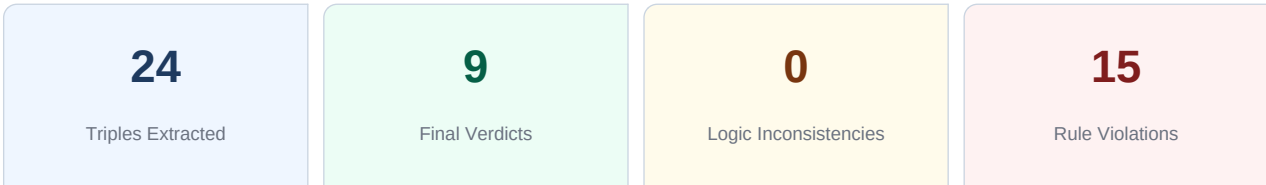
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REPORT OVERVIEW & METHODOLOGY

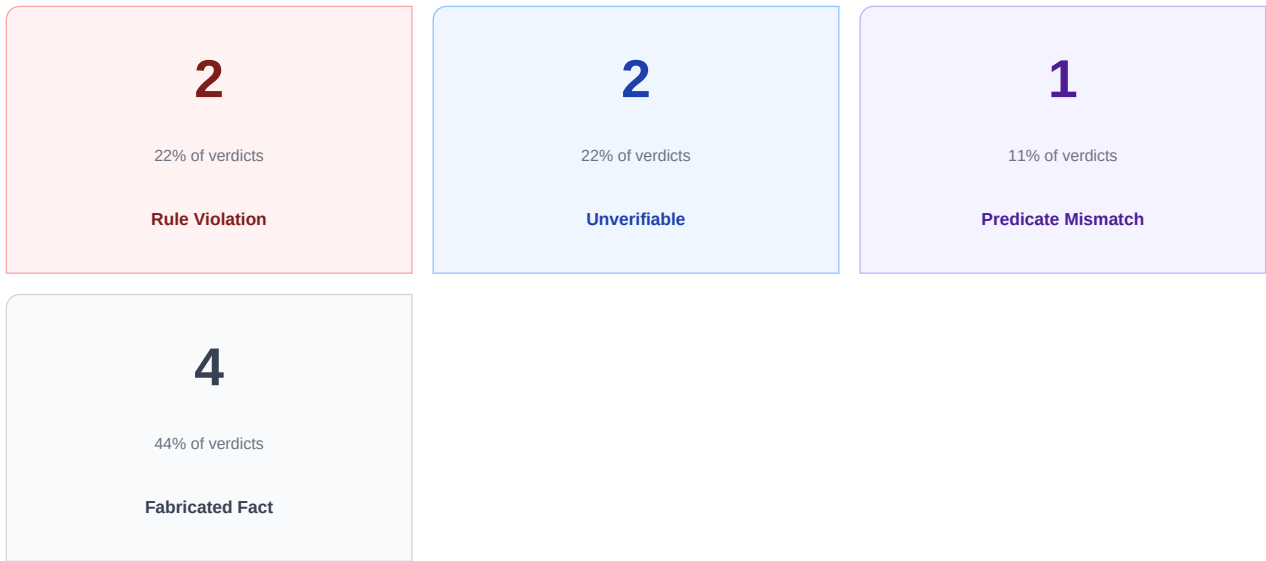
This document outlines the results of the KG-TripleGuard verification pipeline, a neuro-symbolic approach designed to enforce domain rules and verify factual consistency in Black-Box Large Language Models (LLMs).

Pipeline Stages:

1. **Triple Extraction & Canonicalization:** Unstructured LLM outputs are decomposed into relational triples (Subject, Predicate, Object) and mapped to the formal schema of the underlying database.
2. **Symbolic Constraint Verification:** Triples are subjected to rigorous logical constraint checks to identify domain-rule violations prior to graph traversal.
3. **Graph-based Factual Verification:** Canonicalized triples are queried against the Neo4j Knowledge Graph to categorize them as strictly correct, unverifiable, or explicitly fabricated.



VERDICT DISTRIBUTION



EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response prior to canonicalization or knowledge-graph checking.

#	Subject	Predicate	Object	Type	Predicate Definition
1	890	HAS_INTEREST_RATE	30		The interest rate associated with the subject.
2	290	HAS_INTEREST_RATE	20		The interest rate associated with the subject.
3	333	HAS_INTEREST_RATE	40		The interest rate associated with the subject.
4	333	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
5	598	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
6	633	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
7	729	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
8	833	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
9	890	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
10	290	HAS_RISK_TIER	high prime		The risk category or tier assigned to the subject.
11	301	HAS_RISK_TIER	deep subprime		The risk category or tier assigned to the subject.
12	333	HAS_APPROVAL_ODDS	200		The likelihood or odds of approval for the subject.
13	290	HAS_APPROVAL_ODDS	30		The likelihood or odds of approval for the subject.
14	890	HAS_APPROVAL_ODDS	40		The likelihood or odds of approval for the subject.
15	300	HAS_INTEREST_RATE	2		The interest rate associated with the subject.
16	303	HAS_INTEREST_RATE	3		The interest rate associated with the subject.
17	333	HAS_INTEREST_RATE	4		The interest rate associated with the subject.
18	777	HAS_INTEREST_RATE	6		The interest rate associated with the subject.
19	loan_100000_usd	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, collateral, or a certain debt-to-income ratio rather than a fixed account balance.		The account balance required for the subject.
20	300	HAS_DESCRIPTION	Deep subprime (as implied by the risk tier of 301 being 'deep subprime' and the very low approval odds). This score indicates a high-risk borrower with significant credit issues.		A textual description or explanation of the subject.
21	300	HAS_APPROVAL_ODDS	1		The likelihood or odds of approval for the subject.
22	305	HAS_APPROVAL_ODDS	2		The likelihood or odds of approval for the subject.

#	Subject	Predicate	Object	Type	Predicate Definition
23	405	HAS_APPROVAL_ODDS	3		The likelihood or odds of approval for the subject.
24	730	HAS_APPROVAL_ODDS	4		The likelihood or odds of approval for the subject.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. The Confidence column reflects alignment quality.

Subject	Original Predicate	Object	Canonical Predicate	Conf.
890	HAS_INTEREST_RATE	30	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
290	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	40	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
598	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
633	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
729	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
833	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
290	HAS_RISK_TIER	high prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
301	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
333	HAS_APPROVAL_ODDS	200	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
290	HAS_APPROVAL_ODDS	30	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
890	HAS_APPROVAL_ODDS	40	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
300	HAS_INTEREST_RATE	2	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
303	HAS_INTEREST_RATE	3	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
333	HAS_INTEREST_RATE	4	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
777	HAS_INTEREST_RATE	6	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM

Subject	Original Predicate	Object	Canonical Predicate	Conf.
loan_100000_usd	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, collateral, or a certain debt-to-income ratio rather than a fixed account balance.	HAS_REQUIRED_ACCOUNT_BALANCE	LOW
300	HAS_DESCRIPTION	Deep subprime (as implied by the risk tier of 301 being 'deep subprime' and the very low approval odds). This score indicates a high-risk borrower with significant credit issues.	HAS_DESCRIPTION	LOW
300	HAS_APPROVAL_ODDS	1	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
305	HAS_APPROVAL_ODDS	2	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
405	HAS_APPROVAL_ODDS	3	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM
730	HAS_APPROVAL_ODDS	4	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	MEDIUM

FACTUALLY INCONSISTENT TRIPLES

All triples labelled factually_inconsistent from the combined verification agent. Part A shows Pass-1 symbolic constraint violations. Part B shows Pass-2 hallucination verdicts with full evidence.

PART A — Symbolic Constraint Violations (Pass 1)

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	30	c_interest_rate_range	The subject (credit score) 890 violates the constraint because it is greater than 850.
factually_inconsistent	RULE_VIOLATION	290	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	20	c_interest_rate_range	The subject (credit score) 290 violates the constraint because it is less than 300.
factually_inconsistent	RULE_VIOLATION	333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	40	c_interest_rate_range	The object (interest rate) 40 violates the constraint because it is greater than 30.
factually_inconsistent	RULE_VIOLATION	333	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_300_579	The object (risk tier) 'high prime' does not semantically contain 'deep subprime' for subject 333.
factually_inconsistent	RULE_VIOLATION	598	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_580_619	The object (risk tier) 'high prime' does not semantically contain 'subprime' for subject 598.
factually_inconsistent	RULE_VIOLATION	633	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_620_659	The object (risk tier) 'high prime' does not semantically contain 'near prime' for subject 633.
factually_inconsistent	RULE_VIOLATION	729	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_660_739	The object (risk tier) 'high prime' does not semantically contain 'prime' for subject 729.
factually_inconsistent	RULE_VIOLATION	833	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_740_850	The object (risk tier) 'high prime' does not semantically contain 'super prime' for subject 833.
factually_inconsistent	RULE_VIOLATION	890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_740_850	The object (risk tier) 'high prime' does not semantically contain 'super prime' for subject 890.
factually_inconsistent	RULE_VIOLATION	290	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	high prime	c_tier_300_579	The object (risk tier) 'high prime' does not semantically contain 'deep subprime' for subject 290.
factually_inconsistent	RULE_VIOLATION	333	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	200	c_approval_odds_range	The object (approval odds) 200 violates the constraint as it exceeds the maximum allowed value of 100.
factually_inconsistent	RULE_VIOLATION	290	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	30	c_approval_odds_range	The subject (credit score) 290 violates the constraint as it is below the minimum allowed value of 300.

Label	Inconsistency Type	Subject	Predicate	Object	Constraint ID	Explanation
factually_inconsistent	RULE_VIOLATION	890	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	40	<i>c_approval_odds_range</i>	The subject (credit score) 890 violates the constraint as it exceeds the maximum allowed value of 850.
factually_inconsistent	RULE_VIOLATION	405	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	3	<i>c_approval_odds_range</i>	The object (approval odds) value 3 is outside the valid range of 0-100.
factually_inconsistent	RULE_VIOLATION	730	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	4	<i>c_approval_odds_range</i>	The object (approval odds) value 4 is outside the valid range of 0-100.

PART B — Hallucination Detection Verdicts (Pass 2)

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
301	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	Fabricated Fact	factually_inconsistent	FABRICATED_D_FACT	NONE	SUBJECT	0.60	The subject '301' does not exist in the DB candidates for the given predicate and object. Ev: subject: 375, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 380, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 385, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 390, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 395, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime
300	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	2	Rule Violation	factually_inconsistent	RULE_VIOLATION	STRICT	OBJECT	1.00	The object '2' contradicts the stored value '22.94' for the given subject and predicate. Ev: subject: 300, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.94

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
303	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	3	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	0.20	The predicate is known, but the subject and object are absent from the DB.
333	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	4	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	0.20	The predicate is known, but the subject and object are absent from the DB.
777	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	6	Fabricated Fact	factually_inconsistent	FABRICATED_FACT	NONE	MULTIPLE	0.20	The predicate is known, but the subject and object are absent from the DB.
loan_100000_usd	HAS_REQUIRED_ACCOUNT_BALANCE	Not explicitly provided in the given context. Typically, lenders require proof of income, collateral, or a certain debt-to-income ratio rather than a fixed account balance.	Unverifiable	unverifiable	—	NONE	—	0.00	No DB overlap for any component and predicate is unknown.
300	HAS_DESCRIPTION	Deep subprime (as implied by the risk tier of 301 being 'deep subprime' and the very low approval odds). This score indicates a high-risk borrower with significant credit issues.	Unverifiable	unverifiable	—	NONE	—	0.00	No DB overlap for any component and predicate is unknown.
300	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	1	Rule Violation	factually_inconsistent	RULE_VIOLATION	STRICT	OBJECT	1.00	The object '1' contradicts the stored value '8' for the given subject and predicate. Ev: subject: 300, predicate: APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD, object: 8

Subject	Predicate	Object	Verdict	Label	Inconsistency Type	Match	Violated	Conf.	Explanation
305	APPROVAL_ODDS_ESTIMATED_AS_LENDER_ACCEPTANCE_LIKELIHOOD	2	Predicate Mismatch	factually_inconsistent	PREDICATE_MISMATCH	NONE	PREDICATE	0.60	The subject '305' exists, but the predicate is not a valid relation for it in the DB. <i>Ev: subject: 305, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 305, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 22.61</i>

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of any external knowledge source.

No logical inconsistencies detected.

VERDICT LEGEND

Verdict	Description
Correct	All three triple components verified against the knowledge graph.
Rule Violation	One or more components contradict database evidence.
Multi-Rule Violation	Violation spans multiple knowledge-graph triples simultaneously.
Unverifiable	No usable DB evidence found for any component of this triple.
Predicate Mismatch	Subject entity exists, but the stated predicate is not a valid relation.
Fabricated Fact	Triple asserts a relationship absent from the knowledge graph entirely.